

INDU 211: INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

Official Term Paper Group Proposal

Department of Mechanical, Industrial and Aerospace Engineering · Concordia University

Term: Fall 2026 · Course & Section: INDU 211 / Section 2X · Due Date: Thursday, October 8, 2026

1. TEAM IDENTIFICATION AND MEMBER RESPONSIBILITIES

- **Team Name:** Team 4 — Production and Systems Engineering Group
- **Team Members & Work Breakdown:**
 - **Student 1 (Project Coordinator & Systems Lead)** — ID: 40XXXXXX — Overall project coordination, introduction, and systems engineering framework (conversion process, feedback loops, decision hierarchies).
 - **Student 2 (AI Fundamentals & Workplace Researcher)** — ID: 40XXXXXX — Traditional predictive machine learning vs. Generative AI taxonomy, technical foundation, and operational workplace transformation.
 - **Student 3 (Production Operations & Scheduling Analyst)** — ID: 40XXXXXX — Automation potential in production planning, machine scheduling, facility layout design, and assembly line balancing.
 - **Student 4 (Quality, Inventory & Ergonomics Analyst)** — ID: 40XXXXXX — Automation potential in inventory control (EOQ/JIT), statistical process control (SPC) defect detection, and ergonomic safety.
 - **Student 5 (Risk, Reliability & Engineering Ethics Lead)** — ID: 40XXXXXX — Technical hallucinations, system errors, black-box opacity, cybersecurity, and OIQ professional legal liability.
 - **Student 6 (Human Factors, Leadership & Education Specialist)** — ID: 40XXXXXX — Human-in-the-Loop engineering integration, shop-floor crisis leadership, and undergraduate curriculum skills.

2. PAPER TITLE

The Future of Industrial Engineering in the Generative AI Era: Opportunities, Limitations, and the Role of Human Engineers

3. PROJECT SCOPE, OBJECTIVES, AND GROUNDING IN INDU 211 CURRICULUM

Industrial Engineering (IE) focuses on designing, analyzing, and optimizing the integrated transformation systems that bring together people, machines, materials, information, and energy. Our term paper investigates how Generative AI (GenAI) is reshaping these systems.

Rather than treating AI as a generic IT topic, our study is grounded directly in the foundational principles of INDU 211: the **Conversion Process Model**, **Open-Loop vs. Closed-Loop** feedback systems, **Decision Hierarchies** (Strategic, Tactical, Control), and core manufacturing operations. The paper specifically addresses the five mandatory questions assigned by the instructor:

1. **AI and GenAI Fundamentals:** Differentiating predictive Machine Learning (numerical classification and regression) from Generative AI (LLMs producing text, code, and CAD layouts), and examining their operational impact on manufacturing environments.
2. **Automation Potential in IE Domains:** Systematically analyzing six core INDU 211 functional areas: production scheduling, facility layout planning, assembly line balancing, inventory control, quality inspection, and workplace ergonomics.
3. **Operational Errors and Vulnerabilities:** Evaluating critical failure modes including AI hallucinations in technical specifications, lack of physical spatial intuition, black-box unexplainability, data privacy risks, and the legal accountability of engineers under the *Ordre des ingénieurs du Québec* (OIQ).
4. **The Indispensable Human Engineer:** Establishing why human engineers remain irreplaceable for shop-floor crisis management, physical problem-solving, cross-functional leadership, and ethical oversight (the Human-in-the-Loop approach).
5. **Essential Competencies for Future Graduates:** Outlining a practical educational blueprint for undergraduate engineers, combining practical data literacy (basic Python and data handling) with core engineering problem-solving and strong interpersonal skills.

4. PROPOSED 10-PAGE TERM PAPER STRUCTURE

1. **Cover Page & Executive Summary:** Official institutional title page and structured executive summary.
2. **1.0 Introduction & Theoretical Framework:** Systems engineering foundation, conversion model, and decision hierarchy.
3. **2.0 Understanding AI and Generative AI in Production:** Technical comparison of predictive vs. generative architectures.
4. **3.0 Functional Analysis of Core IE Areas:** Deep dive into scheduling, facility layout, line balancing, inventory, SPC, and ergonomics with a comprehensive automation matrix.
5. **4.0 Critical Risks, Limitations, and Failure Modes:** Technical hallucinations, common sense deficits, black-box opacity, and OIQ legal ethics.
6. **5.0 The Critical Role of the Human Industrial Engineer:** Systems synthesis, crisis management, and the Human-in-the-Loop model.
7. **6.0 Educational Blueprint for Future IE Graduates:** Technical literacy, prompt auditing, classical math, and leadership.
8. **7.0 Conclusion & 8.0 Academic References:** Final synthesis and peer-reviewed literature citations.

5. PROJECT TIMELINE AND KEY MILESTONES

- **Milestone 1 (Oct 8, 2026):** Submission of Official 1-Page Group Proposal on Moodle.
- **Milestone 2 (Nov 15, 2026):** Literature review completed and synthesis of IE functional automation matrix.
- **Milestone 3 (Dec 1, 2026):** Submission of 15-Minute Group Presentation Video (Zoom recording with equal participation).
- **Milestone 4 (Final Exam Period):** Submission of Complete 10-Page Final Master Report on Moodle.

6. PRELIMINARY ACADEMIC REFERENCES

1. Bányai, T., Veres, C., & Illés, B. (2024). Generative artificial intelligence in smart manufacturing and logistics: A review. *Journal of Manufacturing Systems*, 74, 312–329.
2. Groover, M. P. (2020). *Fundamentals of Modern Manufacturing: Materials, Processes, and Systems* (7th ed.). John Wiley & Sons.
3. Lee, J., & Azamfar, M. (2024). Industrial AI and human-machine collaboration in production systems. *Computers & Industrial Engineering*, 188, 109880.
4. Tao, F., Zhang, M., & Nee, A. Y. (2023). Digital twins and smart manufacturing in modern industry. *IEEE Transactions on Industrial Informatics*, 19(11), 10842–10853.
5. Institute of Industrial and Systems Engineers (IISE). (2025). The evolving role of the industrial engineer in the era of artificial intelligence. *IISE Solutions*, 57(2), 24–31.